Housing Price Prediction Using Multiple Linear Regression

This report presents the development and evaluation of a multiple linear regression model to predict housing prices using socio-economic and structural variables.

1. Data Loading and Initial Exploration

In this section, we load the housing dataset and explore its structure.

```
# Load dataset
housing <- read.csv("housing.csv")

# View first rows
head(housing)
```

```
##      CRIM ZN INDUS CHAS   NOX    RM  AGE    DIS RAD TAX PRATIO LSTAT class
## 1 0.00632 18  3.81    0 0.538 9.075 65.2 4.0900   1 296   15.3  4.98  24.0
## 2 0.11731  0  8.67    0 0.469 8.621 78.9 4.9671   2 242   17.8  9.14  21.6
## 3 0.02729  0  8.57    0 0.469 9.285 61.1 4.9671   2 242   17.8  4.03  34.7
## 4 0.10237  0  3.48    0 0.458 9.598 45.8 6.0622   3 222   18.7  2.94  33.4
## 5 0.09905  0  3.48    0 0.458 9.247 54.2 6.0622   3 222   18.7  5.33  36.2
## 6 0.08985  0  3.58    0 0.458 8.730 58.7 6.0622   3 222   18.7  5.21  28.7
```

```
# Check structure
str(housing)
```

```
## 'data.frame':    513 obs. of  13 variables:
##  $ CRIM  : num  0.00632 0.11731 0.02729 0.10237 0.09905 ...
##  $ ZN    : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
##  $ INDUS : num  3.81 8.67 8.57 3.48 3.48 3.58 9.67 9.07 9.87 9.57 ...
##  $ CHAS  : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ NOX   : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
##  $ RM    : num  9.07 8.62 9.29 9.6 9.25 ...
##  $ AGE   : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
##  $ DIS   : num  4.09 4.97 4.97 6.06 6.06 ...
##  $ RAD   : int  1 2 2 3 3 3 5 5 5 5 ...
##  $ TAX   : int  296 242 242 222 222 222 311 311 311 311 ...
##  $ PRATIO: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
##  $ LSTAT : num  4.98 9.14 4.03 2.94 5.33 ...
##  $ class : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

```
# Summary stats
summary(housing)
```

```
##       CRIM                ZN             INDUS            CHAS
##  Min.   : 0.00632   Min.   :  0.00   Min.   : 1.56   Min.   :0.00000
##  1st Qu.: 0.12308   1st Qu.:  0.00   1st Qu.: 6.73   1st Qu.:0.00000
##  Median : 0.33169   Median :  0.00   Median :11.00   Median :0.00000
##  Mean   : 3.68953   Mean   : 11.21   Mean   :12.72   Mean   :0.06823
##  3rd Qu.: 3.86498   3rd Qu.: 12.50   3rd Qu.:19.50   3rd Qu.:0.00000
##  Max.   :89.03620   Max.   :100.00   Max.   :29.64   Max.   :1.00000
##       NOX               RM              AGE              DIS
##  Min.   :0.4580   Min.   : 6.161   Min.   :  2.90   Min.   : 1.130
##  1st Qu.:0.5240   1st Qu.: 8.283   1st Qu.: 45.40   1st Qu.: 2.104
##  Median :0.5240   Median : 8.704   Median : 77.70   Median : 3.183
##  Mean   :0.5234   Mean   : 8.768   Mean   : 68.72   Mean   : 3.783
##  3rd Qu.:0.5240   3rd Qu.: 9.227   3rd Qu.: 94.10   3rd Qu.: 5.118
##  Max.   :0.5380   Max.   :11.504   Max.   :100.00   Max.   :12.127
##       RAD              TAX            PRATIO          LSTAT
##  Min.   : 1.000   Min.   :187.0   Min.   :12.60   Min.   : 1.73
##  1st Qu.: 4.000   1st Qu.:280.0   1st Qu.:17.40   1st Qu.: 7.12
##  Median : 5.000   Median :334.0   Median :19.10   Median :11.45
##  Mean   : 9.747   Mean   :411.8   Mean   :18.48   Mean   :12.69
##  3rd Qu.:24.000   3rd Qu.:666.0   3rd Qu.:20.20   3rd Qu.:17.09
##  Max.   :24.000   Max.   :711.0   Max.   :22.00   Max.   :37.97
##      class
##  Min.   : 5.00
##  1st Qu.:16.80
##  Median :21.10
##  Mean   :22.47
##  3rd Qu.:25.00
##  Max.   :50.00
```

2. Problem Definition and Variable Selection

The objective of this analysis is to develop a regression model that predicts housing prices based on various socio-economic and structural factors.

The target variable selected for this model is **class**, which represents the median value of owner-occupied homes.

Several predictor variables were considered, including RM, LSTAT, PRATIO, TAX, and CRIM, because these variables are logically related to housing value and are commonly used in housing price analysis.

3. Data Cleaning and Preprocessing

In this section, we check for missing values and prepare the dataset for regression analysis.

```
# Check for missing values
colSums(is.na(housing))
```

```
##   CRIM     ZN  INDUS   CHAS    NOX     RM    AGE    DIS    RAD    TAX PRATIO
##      0      0      0      0      0      0      0      0      0      0      0
##  LSTAT  class
##      0      0
```

```
# Check for duplicates
sum(duplicated(housing))
```

```
## [1] 7
```

```
# Basic structure check again
str(housing)
```

```
## 'data.frame':    513 obs. of  13 variables:
##  $ CRIM  : num  0.00632 0.11731 0.02729 0.10237 0.09905 ...
##  $ ZN    : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
##  $ INDUS : num  3.81 8.67 8.57 3.48 3.48 3.58 9.67 9.07 9.87 9.57 ...
##  $ CHAS  : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ NOX   : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
##  $ RM    : num  9.07 8.62 9.29 9.6 9.25 ...
##  $ AGE   : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
##  $ DIS   : num  4.09 4.97 4.97 6.06 6.06 ...
##  $ RAD   : int  1 2 2 3 3 3 5 5 5 5 ...
##  $ TAX   : int  296 242 242 222 222 222 311 311 311 311 ...
##  $ PRATIO: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
##  $ LSTAT : num  4.98 9.14 4.03 2.94 5.33 ...
##  $ class : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

The dataset was examined for missing values and duplicate records. No missing values were found; however, 7 duplicate observations were identified. These duplicates are relatively small compared to the dataset size and are unlikely to significantly impact the model.

All variables are numeric, which simplifies the modeling process and ensures compatibility with linear regression techniques.

4. Regression Model Development

In this section, a multiple linear regression model is developed to predict housing prices using selected predictor variables.

```
# Build regression model
model <- lm(class ~ RM + LSTAT + PRATIO + TAX + CRIM, data = housing)

# View model summary
summary(model)
```

```
##
## Call:
## lm(formula = class ~ RM + LSTAT + PRATIO + TAX + CRIM, data = housing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.4311  -3.0239  -0.8499   1.9124  30.1713
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 19.917858   4.514753   4.412 1.25e-05 ***
## RM           3.244161   0.381324   8.508  < 2e-16 ***
## LSTAT       -0.617968   0.047200 -13.093  < 2e-16 ***
## PRATIO      -0.944590   0.128668  -7.341 8.48e-13 ***
## TAX         -0.001054   0.001967  -0.536    0.592
## CRIM        -0.044338   0.034897  -1.271    0.204
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.382 on 507 degrees of freedom
## Multiple R-squared:  0.6579, Adjusted R-squared:  0.6545
## F-statistic:   195 on 5 and 507 DF,  p-value: < 2.2e-16
```

The regression model was built using multiple predictor variables that are expected to influence housing prices. The model estimates the relationship between the target variable (`class`) and the selected predictors.

The `summary()` output provides key statistics such as coefficients, p-values, and R-squared, which help evaluate the strength and significance of the model.

5. Model Evaluation

In this section, the performance of the regression model is evaluated using key statistical metrics.

```
# R-squared
summary(model)$r.squared
```

```
## [1] 0.6578843
```

```
# Adjusted R-squared
summary(model)$adj.r.squared
```

```
## [1] 0.6545104
```

```
# RMSE
rmse <- sqrt(mean(residuals(model)^2))
rmse
```

```
## [1] 5.350328
```

```
# AIC
AIC(model)
```

```
## [1] 3190.595
```

The model achieved an R-squared value of approximately 0.658, indicating that about 65.8% of the variation in housing prices is explained by the selected predictors.

The adjusted R-squared value is slightly lower than the R-squared, indicating that the model does not suffer from excessive overfitting and that the selected predictors contribute meaningfully to the model.

The RMSE value represents the average prediction error of the model. A lower RMSE indicates better predictive accuracy.

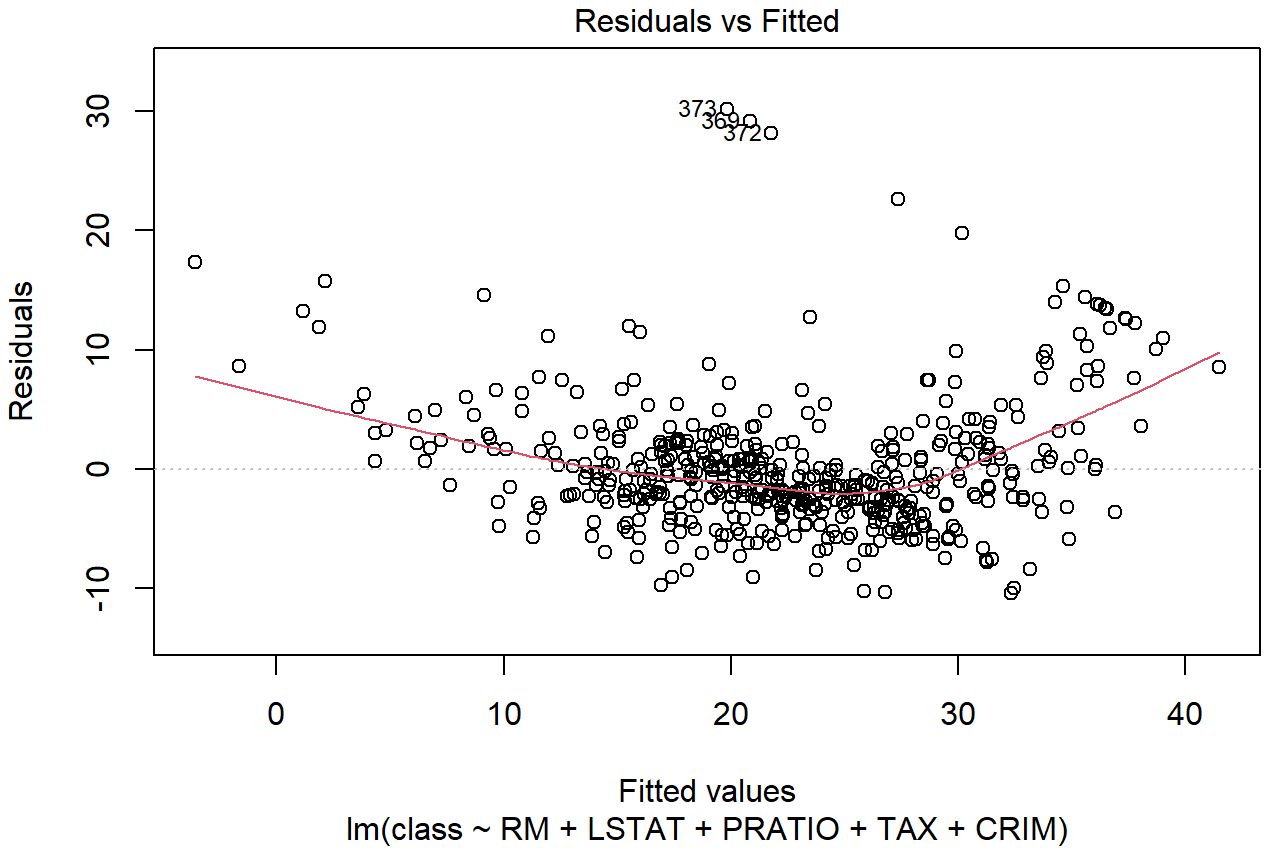
The AIC value provides a measure of model quality while penalizing model complexity. Lower AIC values indicate a better balance between goodness of fit and simplicity.

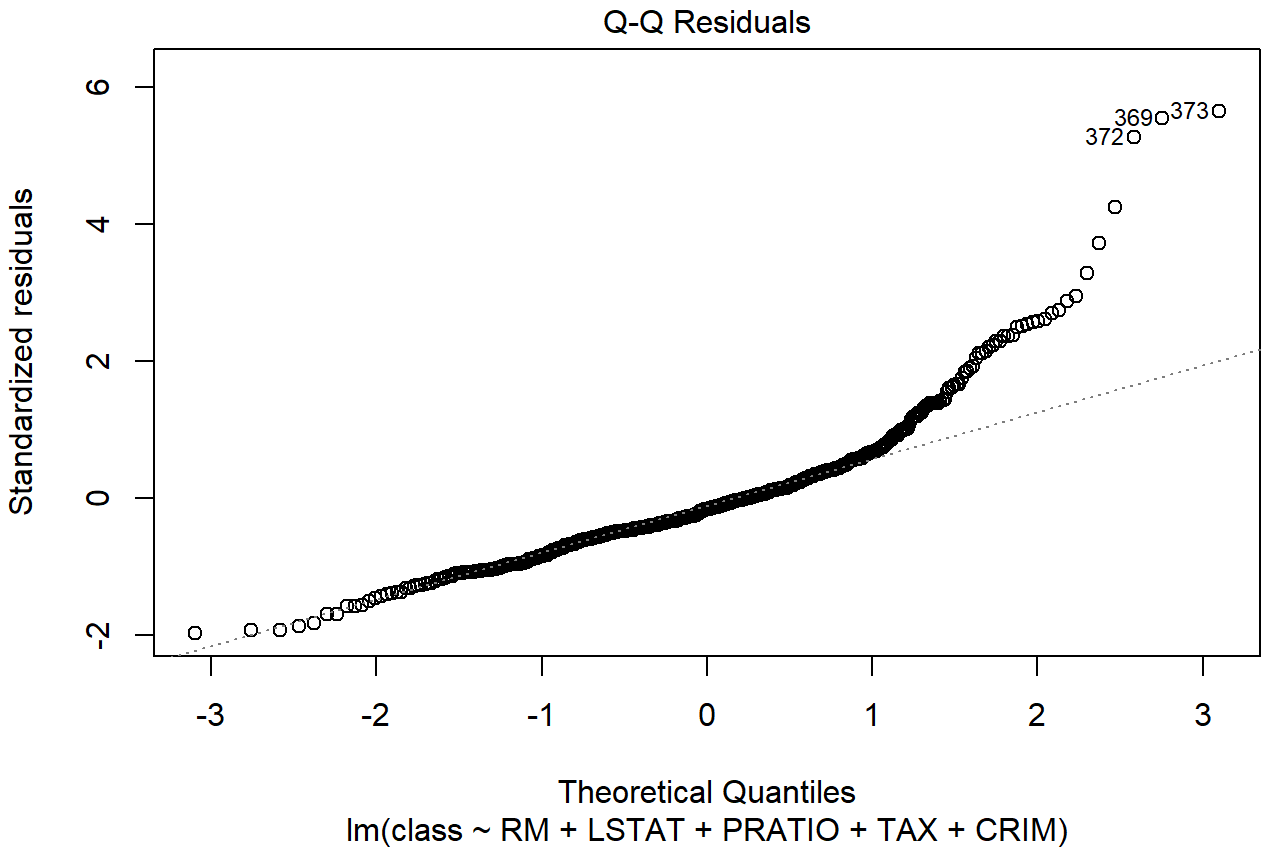
Overall, the model demonstrates a moderately strong fit, explaining a substantial proportion of the variance in housing prices.

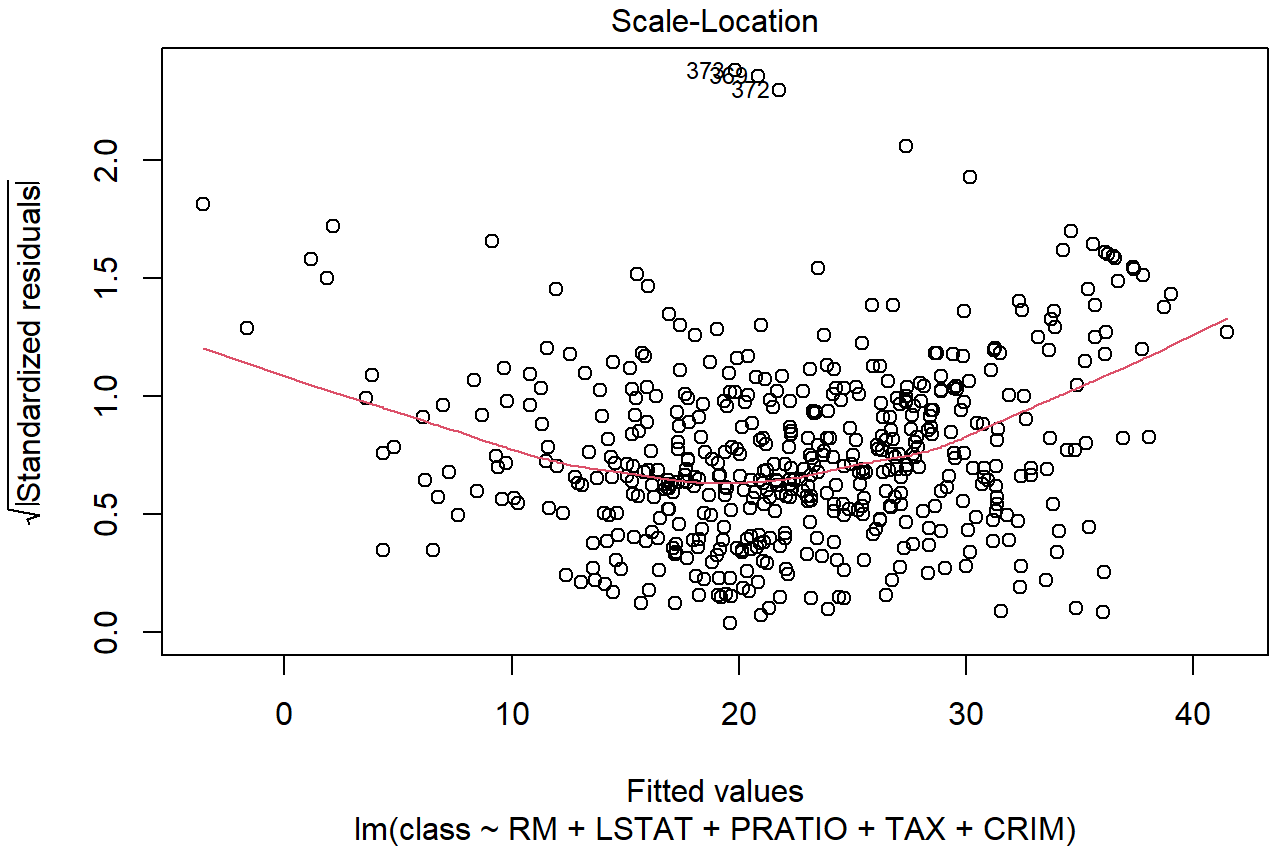
6. Model Diagnostics

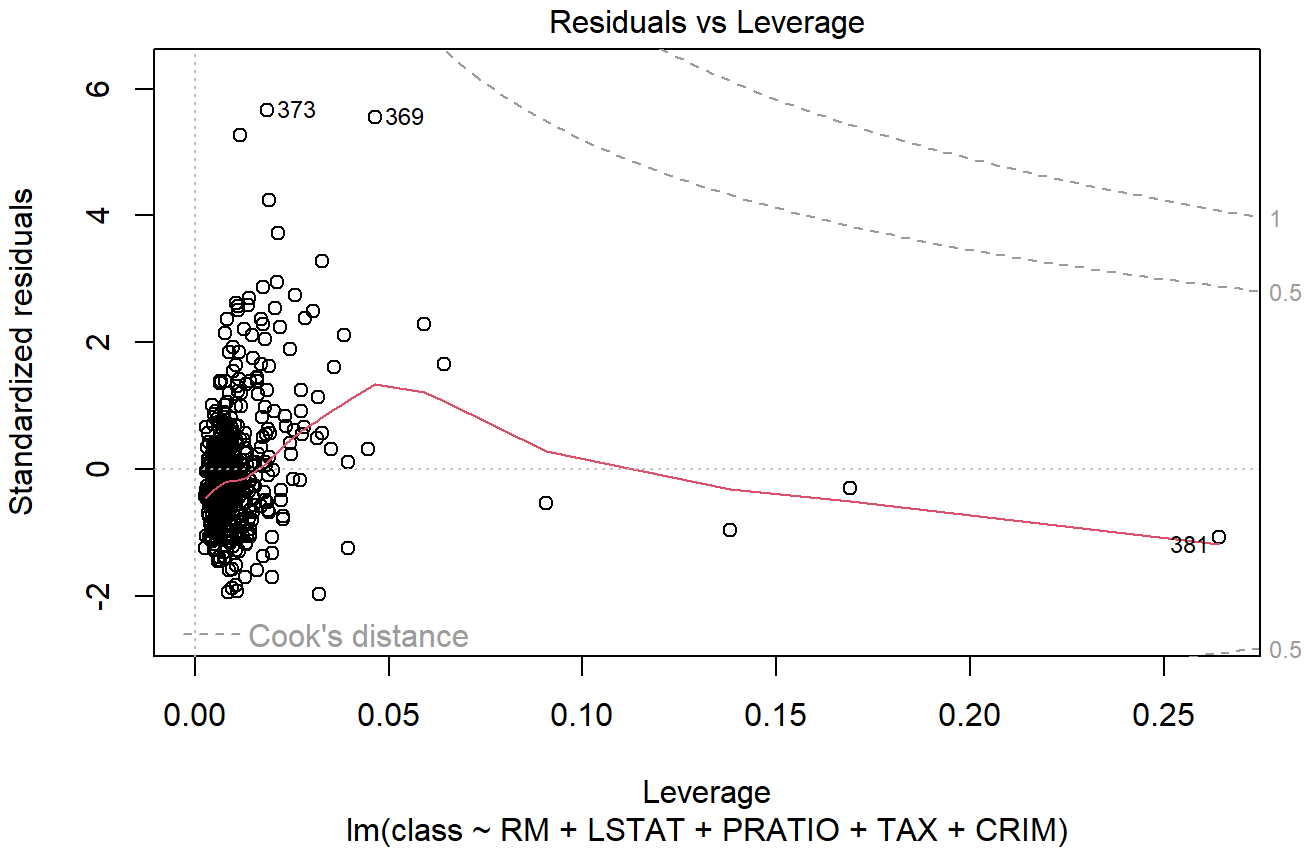
To further evaluate the performance of the regression model, diagnostic plots were generated.

```
# Plot diagnostic graphs
plot(model)
```


Residuals vs Fitted
Residuals
Fitted values
lm(class ~ RM + LSTAT + PRATIO + TAX + CRIM)
373
369
372
30
20
10
0
-10
0
10
20
30
40


Q-Q Residuals
Standardized residuals
Theoretical Quantiles
lm(class ~ RM + LSTAT + PRATIO + TAX + CRIM)
369
373
372
6
4
2
0
-2
-3
-2
-1
0
1
2
3


Scale-Location
√|Standardized residuals|
Fitted values
lm(class ~ RM + LSTAT + PRATIO + TAX + CRIM)


Residuals vs Leverage
Standardized residuals
373
369
381
Cook's distance
Leverage
lm(class ~ RM + LSTAT + PRATIO + TAX + CRIM)

The diagnostic plots provide insights into the validity of the regression assumptions, including linearity, normality of residuals, and homoscedasticity.

These visualizations help identify any potential issues such as non-linearity, outliers, or unequal variance in the residuals.

The diagnostic plots provide additional insight into the assumptions of the regression model.

The **Residuals vs Fitted** plot shows a slight curve, suggesting that the relationship may not be perfectly linear and that some non-linearity may exist in the data.

The **Normal Q-Q plot** indicates that most residuals follow a normal distribution, although there are some deviations at the extremes, suggesting the presence of outliers.

The **Scale-Location plot** shows a slight increase in spread, indicating mild heteroscedasticity (non-constant variance).

The **Residuals vs Leverage plot** identifies a few potential influential observations; however, none appear to strongly dominate the model.

Overall, while the model meets many of the regression assumptions reasonably well, minor deviations suggest that further refinement or transformation of variables could improve model performance.

7. Interpretation of Results

The regression results provide insights into how different factors influence housing prices.

- **RM (number of rooms)** has a positive coefficient (3.24), indicating that houses with more rooms tend to have higher prices.
- **LSTAT (lower status population percentage)** has a strong negative impact (-0.62), suggesting that areas with higher socio-economic disadvantage tend to have lower housing values.
- **PRATIO (pupil-teacher ratio)** also has a negative relationship (-0.94), indicating that poorer educational conditions are associated with lower housing prices.

The variables **TAX** and **CRIM** were not statistically significant ($p > 0.05$), meaning they do not have a strong independent effect in this model.

These findings align with real-world expectations, where housing prices are influenced by property size, socio-economic conditions, and quality of local infrastructure.

Among all predictors, LSTAT appears to have the strongest impact on housing prices, as indicated by its large magnitude and high statistical significance.

8. Conclusion

This analysis successfully developed a multiple linear regression model to predict housing prices using selected socio-economic and structural variables.

The model demonstrated a good level of explanatory power, with key variables such as number of rooms, socio-economic status, and educational conditions significantly influencing housing values.

While the model performs reasonably well, there is still room for improvement. Future work could include adding more predictors, removing insignificant variables, or exploring more advanced modeling techniques.

Overall, the results provide meaningful insights into the factors that drive housing prices.